

BLUESHIELD FILTER: Stakeholder Perceptions and a Preliminary Design-and-Engineering Assessment for Floodwater Treatment

Sabik Bin Sultan¹ Shafi Bin Sultan²

¹Bangladesh Air Force Shaheen College Kurmitola | sabikbinsultan@gmail.com

²St. Joseph Higher Secondary School | shafibinsultan0207@gmail.com

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ABSTRACT

Floodwater-treatment design requires both credible treatment barriers and an understanding of implementation needs. This exploratory cross-sectional study examines stakeholder perceptions of BLUESHIELD FILTER, a proposed system combining coagulation, supported graphene oxide filtration, ultraviolet disinfection and water-quality monitoring. The reported survey comprises 125 students and professionals from Bangladesh, India and Pakistan, with a reported 25-professional interview subsample. Favorable perceptions of technical feasibility, anticipated effectiveness and monitoring usefulness are 84.0%, 86.4% and 92.0%, respectively. Reported concerns include variable source water, energy, affordability and regional suitability. An aggregate audit identifies an unresolved impact-item discrepancy, which is excluded from interpretation. Component evidence informs a treatment sequence with solids separation before the membrane, terminal ultraviolet disinfection and protected storage. Blender visualizations specify stage interfaces and an integrated installation. An illustrative 20-L batch calculation yields 5.71 L/h cycle-average delivery despite an assumed 10 L/h discharge flow; membrane-area, pressure and ultraviolet-exposure calculations identify further prototype requirements. These calculations use stated assumptions and do not establish treatment performance. Undocumented recruitment and instruments, unavailable respondent-level records and the absence of experimental validation limit reproducibility and generalizability. The study supports priorities for documented survey expansion and prototype testing rather than claims of safety, removal efficiency or household adoption.

List of abbreviations

GO	Graphene oxide	UV-C	Ultraviolet C
IoT	Internet of Things	TDS	Total dissolved solids
LMH	Litres per square metre per hour	LRV	Log reduction value
UVT	Ultraviolet transmittance	CFD	Computational fluid dynamics

I. INTRODUCTION

Floodwater treatment presents both an engineering problem and an implementation problem. In Bangladesh, disruption of water supplies can expose communities to contaminated drinking water. Islam et al. examined drinking-water sources during and after the 2004 Dhaka flood and documented faecal

contamination, providing direct evidence for the need to verify microbial quality during flood emergencies [1]. Treatment options must consequently be evaluated against the source water actually encountered and the conditions under which people obtain, store, and use treated water.

Acceptance cannot be inferred from treatment capability

alone. A randomized trial in Dhaka found low use of several household treatment products despite free provision and information [2]. A later field trial comparing automated and manual chlorination showed why sustained operation and user effort deserve attention alongside microbial outcomes [3]. These findings motivate early stakeholder research, while also showing that favorable opinions about a proposed device require subsequent observation of actual use.

BLUESHIELD FILTER is proposed as a modular floodwater-treatment concept that combines particle removal, a graphene-based separation stage, UV-C disinfection, and monitoring. This study asks how a recruited group of students and professionals perceives the concept's feasibility, expected effectiveness, monitoring benefits, and implementation constraints. Its primary contribution is a descriptive survey of those perceptions. A secondary contribution links component evidence to explicit treatment interfaces, three-dimensional design visualizations and a preliminary engineering assessment. The contribution is the survey-informed formulation of testable requirements; uniqueness or superiority of the component combination is not established. No operational prototype performance is inferred from participant ratings.

II. LITERATURE REVIEW

Coagulation provides a relevant pretreatment mechanism when suspended matter compromises downstream treatment. Experimental work on alum-based treatment across different raw-water turbidities illustrates the dependence of operation and sludge production on source-water conditions [4]. Comparative coagulation experiments also identify residual aluminium as an operational consideration [5]. These studies support controlled dosing, mixing, separation of formed flocs, and measurement of residuals; they do not justify prescribing a single alum dose for all floodwaters.

Graphene oxide (GO) offers a platform for engineered separation, but its performance depends on how the material is fabricated and stabilized. Yeh et al. showed that membrane stability in water can depend on multivalent-cation crosslinking [6]. Abraham et al. demonstrated ion sieving through controlled membrane spacing [7], while Qian et al. examined crosslinked GO membranes with ion-permeation and antibacterial measurements [8]. These distinct architectures cannot be treated as interchangeable. More recent porous-GO experiments show that permeance and solute rejection vary with fabrication and operating conditions [9]. The evidence supports selecting and testing a defined membrane or sorbent architecture rather than assigning universal purification properties to graphene.

UV-C effectiveness depends on the radiation delivered to microorganisms within a reactor. Batch and flow-through experiments show that wavelength and reactor configuration affect inactivation [10]. Experiments on turbidity composition further demonstrate that microbial attachment to particles can affect UV performance [11]. Full-scale UV-LED testing using filtered water combined microbial challenges with measurements of UV transmittance and dose [12]. Together, these findings support clarification before UV exposure and reactor-specific verification; electrical lamp power alone does not describe delivered germicidal fluence.

Low-cost monitoring can make operating conditions visible, provided its limitations are understood. A continuous turbidity-monitor study identified calibration and measurement disturbances as practical concerns [13]. An IoT water-monitoring implementation demonstrated integration of several physicochemical sensors [14]. Such instruments can support maintenance and anomaly detection, but pH, turbidity, and conductivity-derived total dissolved solids (TDS) do not directly measure pathogen abundance. The present study therefore examines monitoring as an operational aid, while retaining laboratory water-quality testing as a separate requirement.

III. METHODOLOGY

A. STUDY DESIGN AND PARTICIPANTS

The study is an exploratory cross-sectional stakeholder survey with a reported interview component. The quantitative sample comprises 125 participants: 75 from Bangladesh and 25 each from India and Pakistan. It includes students and professionals in mechanical engineering, computer science engineering, electrical and electronic engineering, civil engineering, and environmental science. The reported survey uses ordered response categories to examine perceptions of treatment feasibility, expected effectiveness, monitoring benefits, and implementation implications. These outcomes are assessments of a concept rather than measured water-treatment endpoints.

Recruitment probabilities, participation and completion rates, item-level missingness, and the wording of the complete instrument are not documented in the available study record. Accordingly, the analysis remains descriptive and does not estimate population-level acceptance or differences between countries. The STROBE reporting framework identifies the recruitment, measurement, participant-flow, and analysis information needed for a reproducible cross-sectional report [15].

Only aggregate summaries are available; participant records, the complete questionnaire, collection dates, recruitment details and consent or ethics documentation have not been recovered. No ethics approval, exemption or consent procedure is asserted, and the original collection procedures cannot be verified from these summaries.

B. INTERVIEW COMPONENT

The reported interview component involves a subsample of 25 professionals and addresses technical, socioeconomic, and regional implementation concerns. Its available outputs are category counts, including agreement totals, rather than participant quotations or a traceable coding framework. The original account describes recorded and transcribed interviews, but the selection procedure, interview guide, coding steps, and link between qualitative material and the plotted counts cannot be independently established. The counts are therefore treated as descriptive summaries of the reported interview component, without claims of thematic saturation or coding reliability. COREQ provides a relevant reporting framework for documenting these procedures [16].

C. DESCRIPTIVE ANALYSIS AND CONSISTENCY AUDIT

Reported percentages and category counts were checked using Python 3.12.14. The audit checked whether demographic counts sum to the stated sample size, recalculated proportions from explicit counts, and combined the two favorable categories for each principal perception item. Reconstructed counts from percentages are presented only as arithmetic equivalents conditional on a denominator of 125. The audit also compared the embedded figures with their narrative descriptions. No participant records were generated, and no weighting, imputation, hypothesis testing, regression, or psychometric reliability analysis was performed. The literature review is a targeted narrative synthesis of relevant empirical studies, rather than a systematic review or meta-analysis. The additional engineering calculations use explicit scenario inputs and dimensional checks; they are separate from the survey analysis. Three-dimensional scenes were modeled and rendered in Blender 4.3, and sensitivity curves were calculated in Python. Rendering is geometric visualization, not computational fluid dynamics or optical simulation.

Focused literature searches targeted empirical coagulation, supported graphene oxide, ultraviolet disinfection, monitoring and household-use studies, together with authoritative engineering guidance. Publisher or archival full texts were used to verify selected claims and figure reuse terms. Selection was based on relevance to component mechanisms and implementation constraints; the search did not estimate completeness or support a systematic-review claim.

IV. SURVEY RESULTS

A. COMPOSITION OF THE SAMPLE

Bangladesh accounts for 75 participants (60.0%); India and Pakistan each account for 25 (20.0%). Students comprise 70 participants (56.0%) and professionals 55 (44.0%). These composition figures describe whose views are represented; they do not establish national representativeness. The country and status distributions are shown in Figs. 1 and 2.

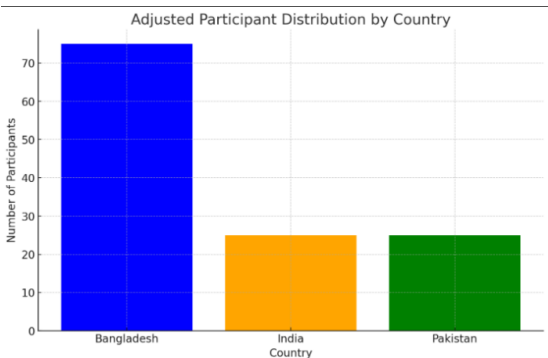


FIGURE 1. Country distribution in the reported survey (n = 125). “Adjusted” is retained from the original graphic; no adjustment or weighting procedure is documented.

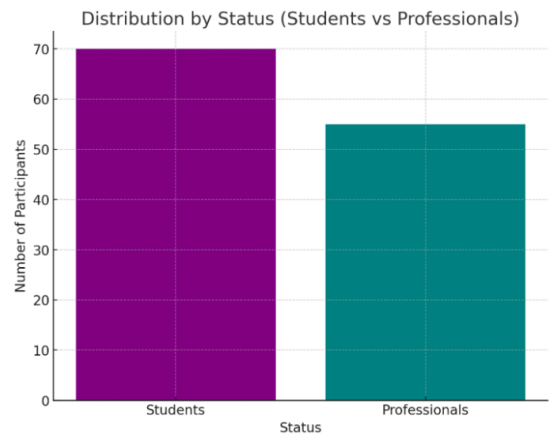


FIGURE 2. Student and professional composition of the reported survey (n = 125). Original survey graphic.

Environmental science and civil engineering contribute 50 and 45 participants, respectively, together representing 76.0% of the sample. Mechanical engineering contributes 10 (8.0%), computer science engineering 12 (9.6%), and electrical and electronic engineering 8 (6.4%). Fig. 3 shows these proportions. The five discipline counts are 8, 10, 12, 45, and 50 when ordered; their median is 12. Fig. 4 summarizes the spread of those five aggregate counts and should not be interpreted as a distribution of participant-level responses.

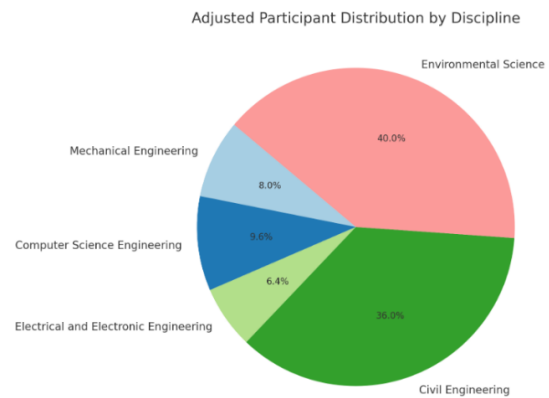


FIGURE 3. Disciplinary composition (n = 125). Original graphic retained; “Adjusted” has no documented analytical definition.

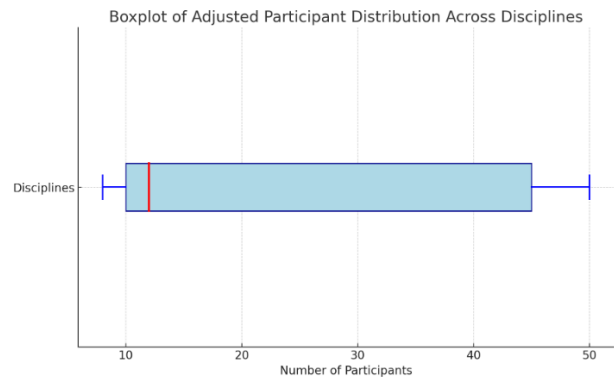


FIGURE 4. Boxplot of the five discipline totals. The median count is 12; the plot does not describe an individual-level outcome.

B. PERCEPTIONS OF THE CONCEPT

The original summary image is retained in Fig. 5 as part of the study record. Experience or interest in flood management or water purification is reported by 57.6%, with 42.4% selecting “No” (Fig. 6). Because experience and interest were grouped, the affirmative proportion must not be read as the proportion with practical expertise.

For technical feasibility, 44.8% selected “somewhat feasible” and 39.2% “very feasible,” giving 84.0% favorable responses (Fig. 7). Anticipated effectiveness received 48.8% “very effective” and 37.6% “somewhat effective,” totaling 86.4% (Fig. 8). Monitoring was considered “very beneficial” by 61.6% and “somewhat beneficial” by 30.4%, totaling 92.0% (Fig. 9). These ratings express expectations about the proposed system; they do not measure removal efficiency, dose delivery, or safety.

Question	Key Aspect	Top Response	Percentage	Insights
1. Prior Experience/Interest	Interest in flood management/purification tech	Yes	57.6%	Majority have relevant experience or interest, forming an informed participant base.
2. Technical Feasibility (IoT Integration)	Feasibility of integrating IoT with purification	Somewhat feasible	44.8%	Indicates optimism tempered by concerns over technical challenges and costs.
3. Effectiveness in Decontaminating Floodwater	Effectiveness of the proposed technology	Very effective	48.8%	Many believe it could significantly remove pathogens and contaminants.
4. IoT-Based Monitoring Benefits	Real-time water quality monitoring	Very beneficial	61.6%	Participants see IoT as essential for proactive decision-making.
5. Social Impact	Social, economic, and environmental benefits	Social (improved health, reduced diseases)	65.6% (Social)	Emphasis on health improvements, affordable clean water, and reduced waste.

FIGURE 5. Original survey summary image. Its social-impact percentage conflicts with Fig. 10; that entry is not used for substantive interpretation.

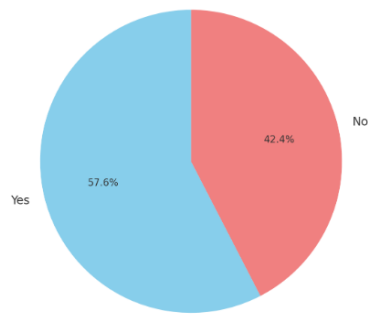


FIGURE 6. Reported prior experience or interest in flood management or water purification (n = 125). Original survey graphic.

The impact item contains an unresolved numerical conflict. The narrative and summary image report social, environmental, and economic percentages of 65.6%, 21.6%, and 12.8%, whereas the corresponding pie chart shows 36.6%, 33.0%, and 30.4% (Fig. 10). The underlying response records and denominator definitions are needed to reconcile these values.

No quantified conclusion about the relative impact domains is drawn from this item.

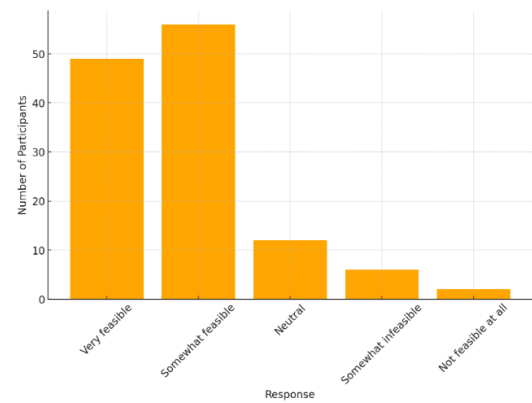


FIGURE 7. Perceived technical feasibility of integrating IoT with UV-C and graphene-based treatment. Original survey graphic.

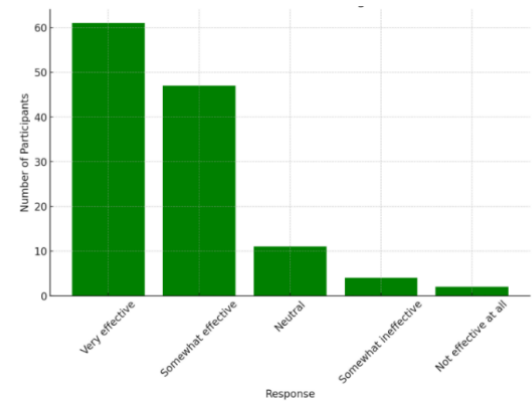


FIGURE 8. Anticipated floodwater-decontamination effectiveness. The bars represent perceptions, not experimental removal percentages.

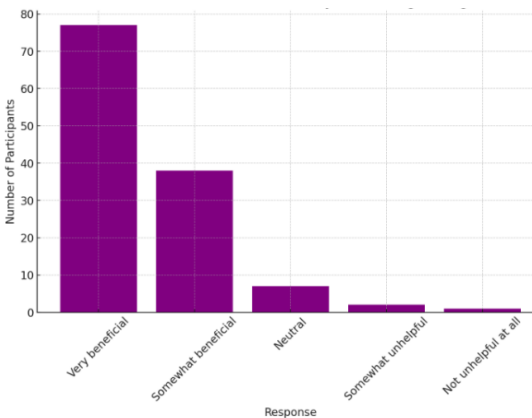


FIGURE 9. Perceived benefit of real-time water-quality monitoring. Original survey graphic.

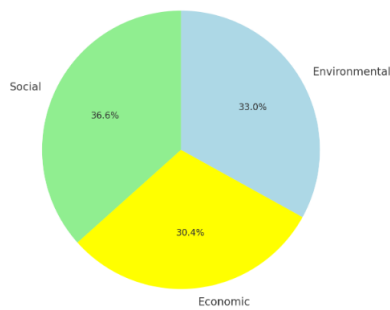


FIGURE 10. Original impact pie chart, retained unchanged. Its displayed percentages conflict with the narrative and summary image; this item is excluded from interpretation pending reconciliation.

C. REPORTED IMPLEMENTATION CONCERNS

Among the 25 professionals in the reported interview component, 20 identified environmental conditions and 15 identified energy efficiency or scalability as concerns (Fig. 11).

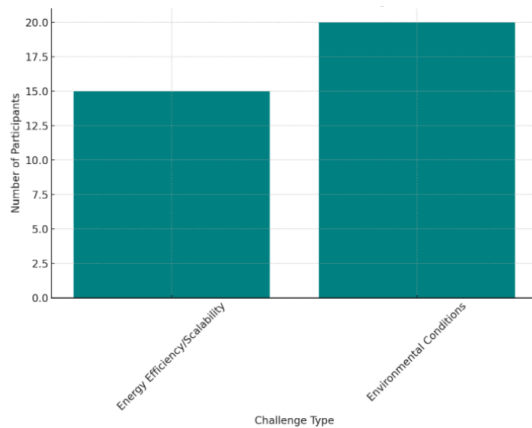


FIGURE 11. Reported technical concerns in the 25-professional component. Categories may overlap; original graphic.

Affordability concerns were reported for 18 participants, while public-health benefits received combined agreement from 22 (Fig. 12).

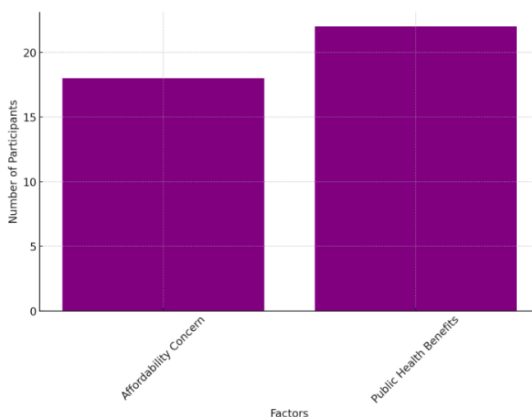


FIGURE 12. Reported affordability concerns and agreement with anticipated public-health benefits (subsample n = 25). Original graphic.

Regional factors received agreement from 23, and cross-border relevance from 22 (Fig. 13). These counts concern

different prompts or categories and are not mutually exclusive shares of a single total. They indicate reported priorities without establishing the prevalence or independent coding of qualitative themes.

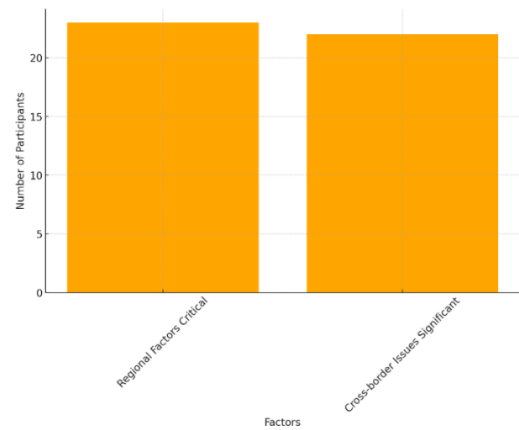


FIGURE 13. Reported agreement with regional and cross-border considerations (subsample n = 25). Original graphic.

D. REPRODUCIBLE AGGREGATE CHECKS

The demographic totals reconcile to 125, and the three combined favorable percentages reconcile arithmetically. Under a common denominator of 125, these percentages correspond to 105 favorable feasibility responses, 108 favorable effectiveness responses, and 115 favorable monitoring responses. These are reconstructed equivalents, not a replacement for an item-level response export. Fig. 14 records the calculation visually. The impact-item conflict remains unresolved; statistical software cannot determine which version is correct without the underlying records.

Reported favorable perceptions

Descriptive aggregate audit; stated survey n = 125

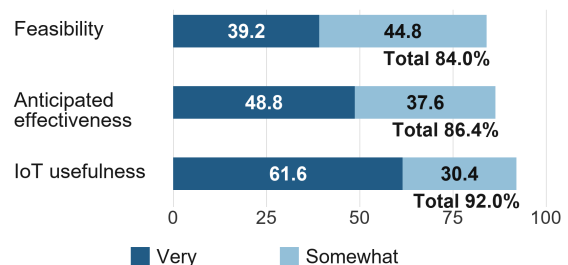


FIGURE 14. Python arithmetic audit of the three reported perception items. Favorable totals combine the reported “very” and “somewhat” categories; bars are descriptive aggregates without inferential uncertainty intervals.

V. BLUESHIELD FILTER CONCEPT

A. TREATMENT SEQUENCE AND PRETREATMENT

The proposed treatment train is intake screening, coagulation and flocculation, settling with sludge removal, a pressure-compatible supported GO-based cartridge, terminal UV-C disinfection, and protected storage. This ordering is an engineering interpretation of the component evidence [4]–[12]. The original component drawings are retained, while the revised sequence places the final disinfection barrier after filtration.

Treatment modules require accessible sampling points so that each stage’s contribution can be measured independently.

In the coagulation module (Fig. 15), alum is a candidate coagulant. The dose and mixing conditions must be established by jar tests using representative source waters, with measurement of pH, alkalinity, clarified-water turbidity, and residual aluminium [4], [5]. Coagulation destabilizes colloidal particles; flocculation promotes growth of aggregates, which settling or another solids-separation step removes. A universal 10–20 mg/L dose and a fixed motor speed are not established design specifications. Sludge removal and cleaning access are necessary design provisions.

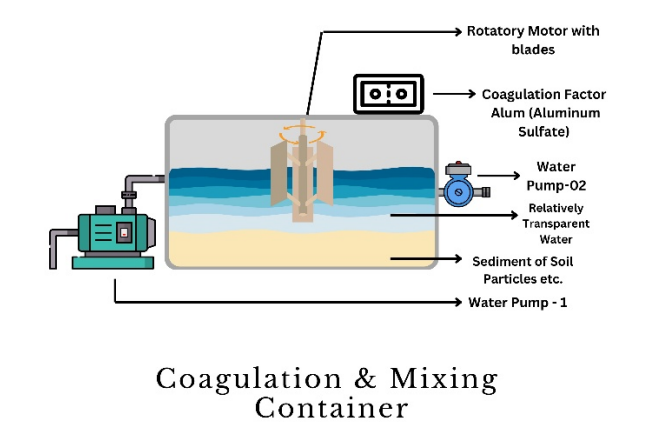


FIGURE 15. Authors’ original Canva drawing of coagulation and mixing, retained unchanged. Equipment dimensions and operating settings remain unvalidated.

Figure 16 supplies an external example of why chemical residuals belong in the pretreatment assessment. Krupińska compared four coagulants using a groundwater–surface-water mixture, with raw-water turbidity 10.6–15.0 NTU and pH 7.60–7.84 [5]. The residual-aluminium pattern varied with coagulant and dose. These conditions do not represent a validated BLUESHIELD floodwater envelope. In particular, the source dose is expressed as elemental Al per litre, whereas the illustrative mass accounting in Section VI uses commercial product mass. The results motivate measuring both clarification and aluminium residuals during jar tests; they do not select a universal coagulant or operating dose.

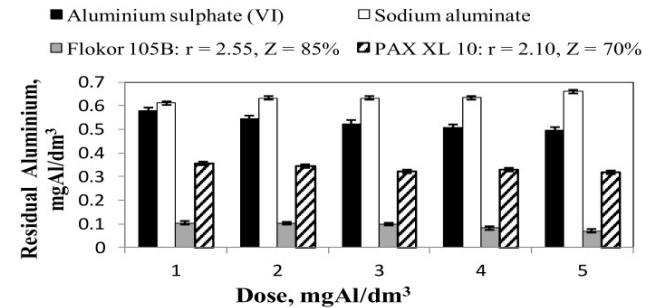


FIGURE 16. External residual-aluminium results versus coagulant dose. The dose axis is mg Al/dm³ (equivalent to mg Al/L), not mg/L commercial product. Source results are means of three jar tests; the plotted error-bar definition is not stated. *r* and *Z* describe coagulant alkalinity ratio and alkalinity percentage. Reproduced unchanged from Krupińska [5], Fig. 3, © 2020 the author, CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>. These are external experiments, not BLUESHIELD measurements.

B. DEFINED GRAPHENE-BASED SEPARATION STAGE

The graphene stage is specified as an immobilized GO composite on a suitable support, rather than loose nanosheets dispersed into product water. Fig. 17 preserves the original vessel concept. Membrane formulation, effective area, thickness, support, pore or interlayer structure, and operating pressure must be selected before flux and selectivity can be evaluated [6]–[9]. A sorbent cartridge would require separate capacity and breakthrough measurements; adsorption and membrane rejection should not be treated as the same mechanism. Fig. 18 provides a spatial illustration of the vessel.

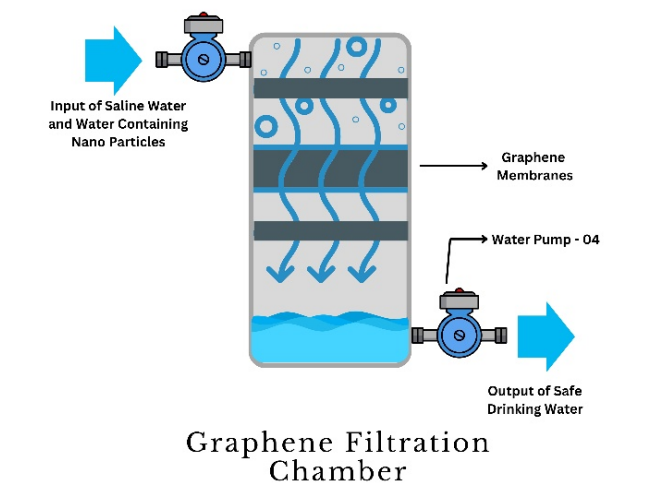


FIGURE 17. Authors’ original Canva drawing of graphene filtration, retained unchanged. The original labels specify a proposed saline feed and intended drinking-water product; neither potability nor desalination has been demonstrated.

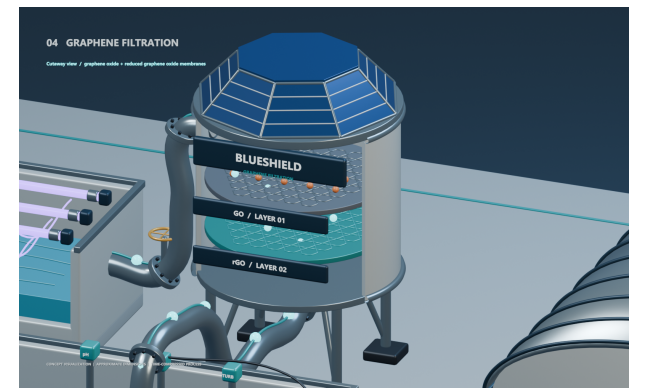


FIGURE 18. Earlier Blender vessel cutaway retained from the project design. Layer arrangement and flow markers are illustrative. The revised integrated design places terminal UV-C after the supported membrane; no transport simulation or validated performance is represented.

Its GO and reduced-GO layers visualize candidate arrangements; they do not establish a fabricated membrane composition. Wet-state integrity, swelling, sealing, fouling, and release of carbonaceous material must be examined under the anticipated operating conditions [6], [7]. Challenge testing should distinguish particulate removal, dissolved-solute rejection, and adsorptive uptake instead of reporting a single “purification efficiency.”

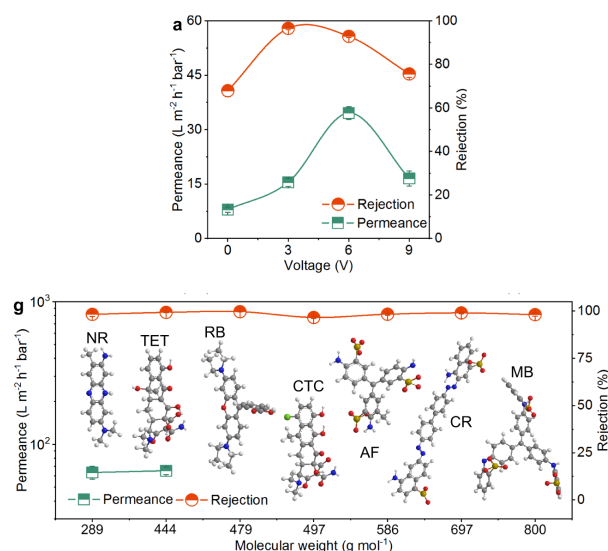


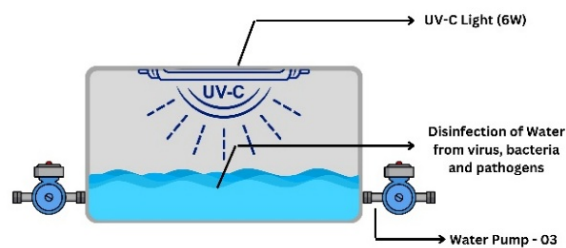
FIGURE 19. External GO results at 2 bar: source Fig. 5(a), 200 μg membranes; Fig. 5(g), 50 μg PGO-6 membrane. PGO denotes porous GO; the suffix in PGO-3, PGO-6 and PGO-9 denotes a fabrication voltage of 3, 6 and 9 V, respectively. Error bars are standard deviations across three membranes. NR, TET, RB, CTC, AF, CR, and MB denote neutral red, tetracycline, rhodamine B, chlortetracycline hydrochloride, acid fuchsin, Congo red, and methyl blue. Reproduced from Liu et al. [9]. © The Author(s) 2024, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Panels excerpted and stacked; data unchanged. These are not BLUESHIELD results.

Published separation results illustrate the need for this specificity. In the porous-GO study by Liu et al., sodium-sulfate separation depended on membrane preparation and test conditions, while sodium-chloride rejection was substantially lower [9]. Fig. 19 reproduces selected experimental panels as external material evidence. These data support testing an engineered separation layer; they cannot be transferred as BLUESHIELD removal rates or as proof of saline-floodwater desalination.

Recent membrane research continues to support architecture-specific evaluation. Han et al. demonstrated that arranging modified GO flakes within an asymmetric porous support altered ion-sieving performance and stability [17]. This provides a further candidate design principle, but its tested metal-salt separations and laboratory operating conditions do not establish general floodwater performance. BLUESHIELD requires its own material characterization and challenge tests before selecting such an architecture.

C. TERMINAL UV-C DISINFECTION

The UV-C chamber (Fig. 20) is a terminal treatment barrier whose performance must be linked to delivered fluence, wavelength, flow distribution, and water UV transmittance [10]–[12]. The original 6 W lamp label is a nominal component rating. It cannot establish a germicidal dose or a microbial log reduction.



UV-C Light Chamber

FIGURE 20. Authors' original Canva UV-C drawing, retained unchanged with its original 6 W label. Electrical wattage does not establish delivered fluence or disinfection performance.

For an idealized uniform field, fluence equals in-water fluence rate multiplied by exposure time. Real reactors require validation that accounts for the least-exposed flow paths, lamp ageing, sleeve fouling, and variation in feed-water transmittance. Particle-associated shielding is a further reason to clarify water before UV [11]. The external dose-response evidence in Fig. 21 illustrates the type of reactor evaluation required [12]. A proposed fail-closed outlet should prevent dispensing during lamp failure, excessive flow, or operation outside a validated treatment envelope. Setpoints remain to be established experimentally.

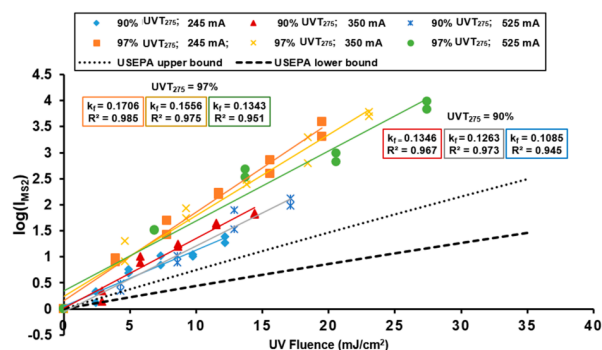


FIGURE 21. External bench-scale UV-LED dose-response for MS2 bacteriophage, compared with mercury-UV reference curves. UVT₂₇₅ denotes UV transmittance at 275 nm. From Jarvis et al. [12], Fig. 4. Reproduced © The Authors 2019 under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>), without changing the plotted results. Source-specific MS2 challenge results do not validate BLUESHIELD.

D. MONITORING AND PROTECTED STORAGE

The proposed sensor package measures pH, temperature, turbidity, and conductivity-derived TDS, with flow and UV status included for process supervision. Calibration against reference instruments, drift checks, cleaning, and local data retention are required because low-cost measurements are sensitive to deployment conditions [13], [14]. The controller should provide local alarms and retain essential functions during loss of connectivity. A dashboard can communicate operating status; it cannot certify microbial or chemical safety from these indicators alone.

Treated water should enter a protected vessel with hygienic dispensing, controlled access, and a documented cleaning pro-

cedure. Storage performance and water quality at the point of use require evaluation, as demonstrated by household field research [2], [3]. Where source-water salinity or a dissolved contaminant exceeds the demonstrated capability of the selected cartridge, the design requires a separately validated treatment option or an alternative water source. Power demand, throughput, consumable life, and cost per treated volume remain evaluation outcomes, not established attributes of the concept.

E. THREE-DIMENSIONAL STAGE VISUALIZATION

The authors’ Canva drawings retain the original design record. New Blender geometries with flat two-dimensional annotations in Figs. 22–24 make the proposed interfaces and maintenance provisions explicit. The models are explanatory, not dimensioned fabrication drawings, manufactured prototypes or

fluid/optical simulations. Blue process paths, separate residuals handling, supported media and an enclosed UV chamber express design intent. Dimensions, materials and operating setpoints require subsequent engineering verification.

Screening and access for debris removal precede rapid mixing and gentle flocculation (Fig. 22). Settling requires a decant outlet separated from accumulated sludge; carrying flocs directly into the GO cartridge would undermine its pretreatment role. The separation models in Fig. 23 depict settling and supported, immobilized membrane media with an inlet, permeate outlet, pressure observation and residuals handling. The GO arrangement does not assert a particular commercial membrane formulation. The terminal-stage models in Fig. 24 place UV treatment after the membrane in a closed chamber with provision for sleeve access and flow supervision. Protected storage and hygienic dispensing complete the train, while local sensing supports operational decisions.

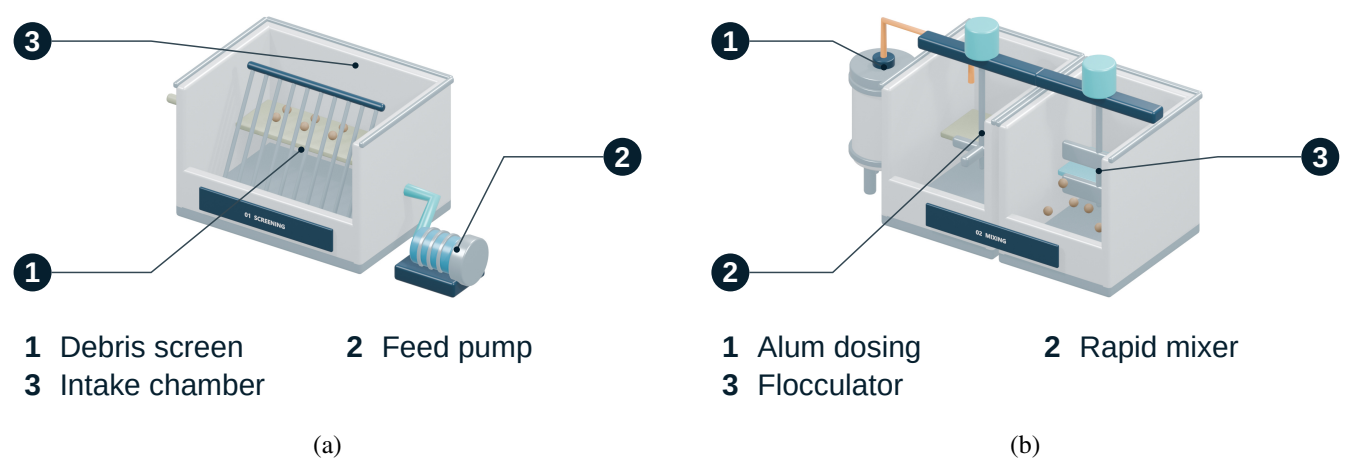


FIGURE 22. Proposed intake and pretreatment in Blender. (a) Screened intake with accessible debris removal. (b) Coagulant addition, rapid mixing and gentle flocculation. Rendered vessels and impellers are conceptual geometries; dose and mixing conditions require jar tests.

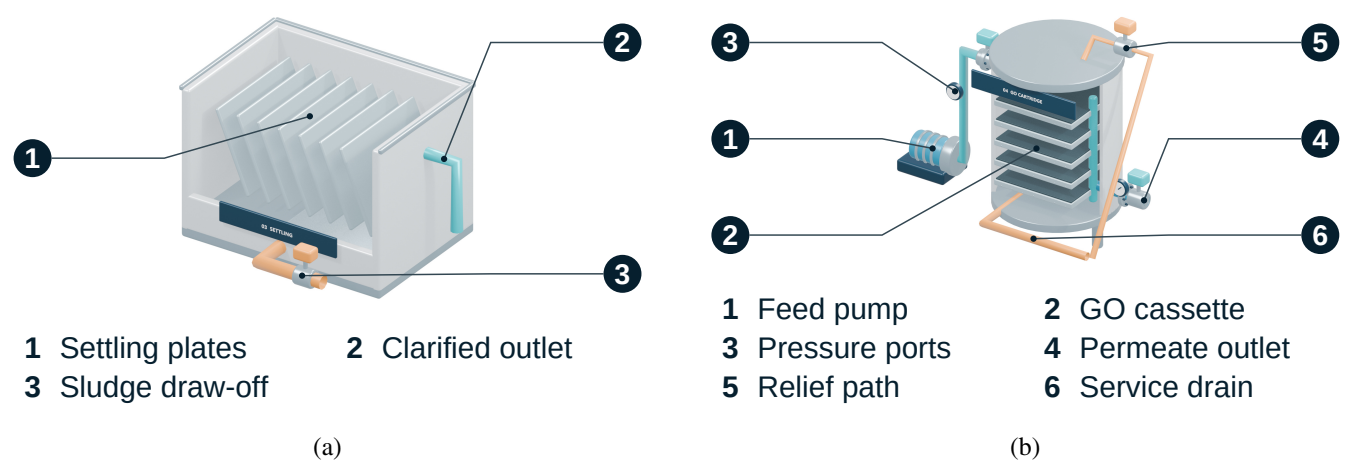
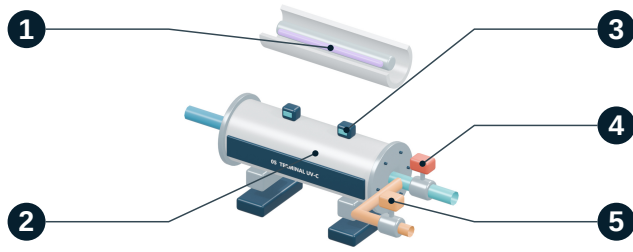
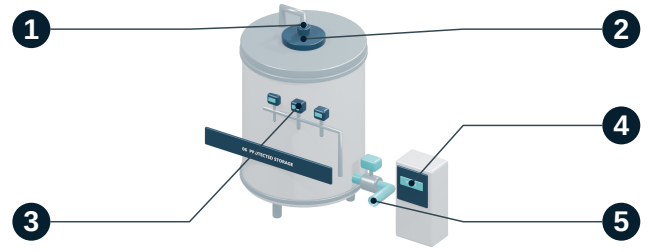


FIGURE 23. Proposed separation stages in Blender. (a) Settling vessel with a distinct sludge withdrawal route. (b) Supported GO cartridge cutaway with contained media and pressure-loop provisions. The cutaway is a structural illustration, not a molecular-scale or pressure-vessel validation.



- 1 Lamp and sleeve
- 2 Closed housing
- 3 Process sensors
- 4 Fail-closed valve
- 5 Fault diversion

(a)



- 1 Protected vent
- 2 Sealed access
- 3 Water sensors
- 4 Status controller
- 5 Hygienic outlet

(b)

FIGURE 24. Proposed terminal barriers in Blender. (a) Enclosed UV-C reactor with an internal explanatory view. (b) Protected storage and monitoring. Final UV follows filtration; sensing and valve control require calibration and functional testing. No water-safety outcome is inferred from the render.

VI. PRELIMINARY ENGINEERING ASSESSMENT

The survey identifies perceived value and implementation concerns; it does not provide hydraulic or treatment parameters. The following calculations translate those concerns into testable design requirements. Every numerical input in Table 1 is an illustrative assumption selected for a small batch unit. None is estimated from respondent ratings or presented as a measured BLUESHIELD specification. These calculations are a screening assessment, without transport simulation, optimization, or experimental validation.

TABLE 1. Illustrative inputs and calculated quantities for a small batch unit. All inputs are assumptions; all outputs are conditional calculations.

Quantity	Value and status
Raw batch volume, V_b	20 L; assumed
Recoverable fraction, f_r	0.90; assumed
Flow during discharge, Q_p	10 L/h; assumed
Rapid mixing / flocculation	1 / 20 min; assumed
Quiescent settling	60 min; assumed
Product volume, V_p	18 L; calculated
Discharge time	1.80 h; calculated
Cycle time	3.15 h; calculated
Cycle-average output	5.71 L/h; calculated

A. BATCH THROUGHPUT AND COAGULANT ACCOUNTING

For sequential mixing, settling and discharge, the volume and time balances are

$$\begin{aligned}
 V_p &= f_r V_b, \\
 t_{\text{cycle}} &= t_{\text{rapid}} + t_{\text{floc}} + t_{\text{settle}} + V_p / Q_p, \\
 \bar{Q}_p &= V_p / t_{\text{cycle}}.
 \end{aligned} \quad (1)$$

The example produces a calculated cycle-average output of 5.71 L/h, even though water is discharged at 10 L/h. Filling, cleaning and operator delays are omitted, so actual average delivery could be lower. The assumed 90% recoverable fraction is a lumped allowance for volume loss; it does not establish membrane recovery, concentrate handling or sustainable operation. A plant with parallel tanks would require a separate scheduling and mass-balance model.

Alum addition can be accounted for without assuming a universally effective dose. For commercial-product dose D in mg/L, batch volume V_b in L and stock concentration C_s in mg/L of the same product,

$$m_a = \frac{DV_b}{1000} [\text{g}], \quad V_s = \frac{DV_b}{C_s} [\text{L}]. \quad (2)$$

Illustrative doses of 10, 30 and 50 mg/L would require 0.20, 0.60 and 1.00 g per 20-L batch, respectively. These are accounting examples, not treatment recommendations. With an assumed 10 000-mg/L stock, the corresponding additions are 20, 60 and 100 mL. Commercial grade, hydrate and assay must be specified because product mass and elemental-aluminium mass are different dose bases. Source-specific jar tests must establish the dose, mixing sequence and settling conditions [18]. Stock additions are neglected in the approximate cycle example and must be included in a final balance.

The bulk velocity-gradient relation is

$$G = \sqrt{\frac{P_w}{\mu V}}, \quad P_w = \mu V G^2, \quad (3)$$

where G is in s^{-1} , P_w is power dissipated into water in W, μ is viscosity in Pa s, and V is water volume in m^3 [19]. Assuming $\mu = 10^{-3}$ Pa s and $V = 0.020$ m^3 , gradients of 700 and 50 s^{-1} imply fluid powers of 9.8 and 0.050 W. Neither value specifies motor demand, impeller speed or local floc shear. Settling effectiveness must likewise be measured; a selected holding time cannot be converted directly to a turbidity-removal percentage.

B. MEMBRANE AREA AND PRESSURE COMPATIBILITY

For a defined supported membrane, operating flux J , apparent applied-pressure permeance $L_{p,\text{app}}$ and solute rejection R_i are

$$J = \frac{Q_p}{A_m}, \quad L_{p,\text{app}} = \frac{J}{\Delta P}, \quad R_i = 1 - \frac{C_{p,i}}{C_{f,i}}. \quad (4)$$

Here J is in $\text{L m}^{-2} \text{h}^{-1}$ (LMH), A_m in m^2 , and feed and permeate concentrations use matching units. ΔP is transmembrane pressure; for cross-flow, it is approximately $(P_f + P_r)/2 - P_p$, using feed, retentate and permeate pressures. It is distinct from

the total pump differential. These definitions are consistent with the controlled experiments in [9]; their reported results are not input performance values for this design. $L_{p,app}$ depends on the solution and operating conditions and is not a universal material constant. Salinity, osmotic effects and concentration polarization require further treatment before a desalination model is possible.

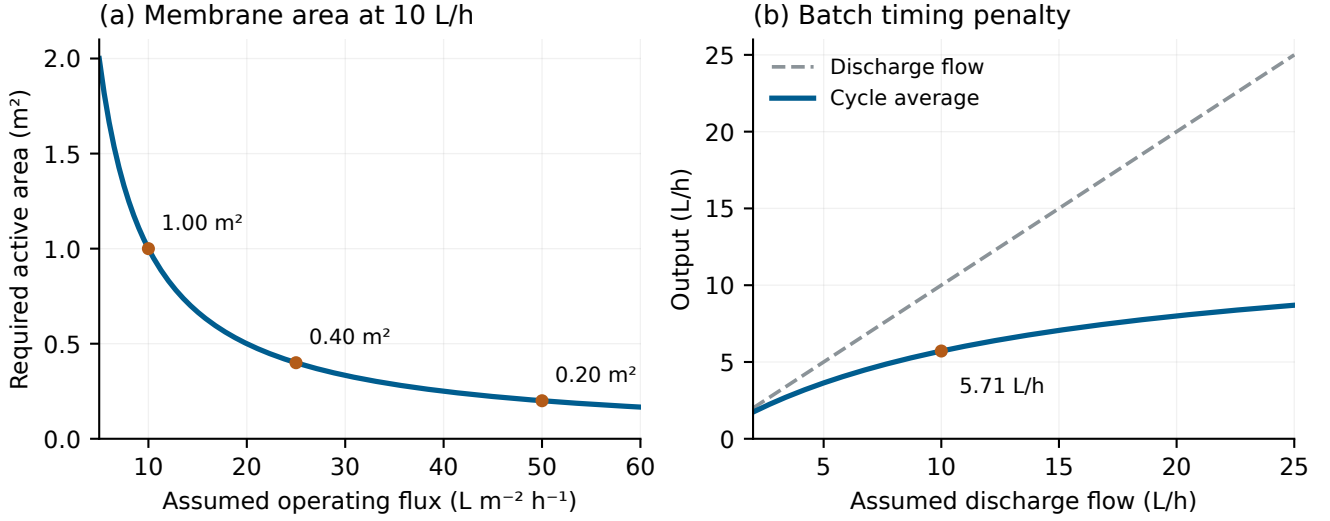


FIGURE 25. Calculated design sensitivities. (a) Active area required for 10 L/h over assumed operating fluxes; the highlighted values are illustrative inputs. (b) Cycle-average output for an assumed 18-L recoverable batch and 81-min combined mixing and settling time, compared with instantaneous discharge flow. These are deterministic calculations, not BLUESHIELD test results; no empirical uncertainty intervals are available.

The available pressure must be compatible with the membrane assumption. Hydrostatic head gives

$$\Delta P = \rho gh. \quad (5)$$

With rounded water density $\rho = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$ and $h = 0.50 \text{ m}$, the head is only 0.0491 bar. An explicitly hypothetical linear extrapolation of 25 LMH/bar would then yield 1.23 LMH and require approximately 8.15 m² to supply 10 L/h. This optimistic illustration neglects additional losses and is not a low-pressure membrane prediction. Conversely, 1 bar corresponds to approximately 10.2 m of water head. A compact gravity vessel therefore cannot inherit flux measured in a pressurized laboratory cell. A pumped concept needs a rated housing, seals, relief provision and a defined reject or cleaning route.

C. UV EXPOSURE AND PUMPING WORK

For a microorganism following trajectory j , delivered fluence is the time integral of the in-water fluence rate [20]:

$$H_j = \int_0^{t_j} E_f(\mathbf{x}_j(t), t) dt. \quad (6)$$

An ideal uniform-field, plug-flow calculation gives

$$H_{\text{nom}} = \bar{E}_f \frac{V_{UV}}{Q}, \quad V_{UV, \text{ideal}} = \frac{QH_{\text{study}}}{\bar{E}_f}. \quad (7)$$

When fluence rate is mW/cm² and fluence is mJ/cm², time must be in seconds. Thus Q in L/h is divided by 3600 before the volume calculation. Table 2 uses an assumed study fluence of 40 mJ/cm²; this is a sensitivity target, not a universal safety

threshold. The volume excludes the lamp sleeve and displaced solids. Real reactors exhibit distributions of exposure, and neither nominal residence time nor electrical wattage establishes a validated reduction-equivalent dose.

TABLE 2. Ideal UV sizing at assumed flow 10 L/h and study fluence 40 mJ/cm². Optical inputs are independent assumptions.

Fluence rate (mW/cm ²)	Exposure (s)	Water volume (L)
0.25	160	0.444
0.50	80	0.222
1.00	40	0.111

The pump energy requirement follows a hydraulic work balance [21]:

$$P_{\text{el}} = \frac{\Delta P_{\text{tot}} Q_{\text{pump}}}{\eta_{\text{sys}}}, \quad e_{\text{pump}} = \frac{\Delta P_{\text{tot}}}{3.6 \times 10^6 \eta_{\text{sys}}}. \quad (8)$$

Pressure is in Pa, pump flow in m³/s, power in W and specific energy in kWh per m³ pumped. With assumed overall efficiency 0.30, total pressure differentials of 0.5 and 2.0 bar give 0.0463 and 0.1852 kWh/m³ pumped. At 10 L/h, the corresponding calculated powers are 0.463 and 1.852 W. Miniature-pump losses, recirculation and the operating point may increase actual draw. These pressures do not imply equal membrane flux. Total treatment energy must additionally include measured mixing, UV, controls, standby and cleaning demand per delivered volume. No relationship is asserted between the original 6-W lamp label and the optical assumptions in Table 2.

D. VALIDATION ENDPOINTS

Removal claims require matched influent and effluent measurements. For the same viable or infective organism and assay basis,

$$\text{LRV}_i = \log_{10} \left(\frac{N_{\text{in},i}}{N_{\text{out},i}} \right), \quad \eta_i = 1 - 10^{-\text{LRV}_i}. \quad (9)$$

A nondetect is bounded by an assay detection limit; it is never entered as zero in the logarithm. Stage log reductions can only be combined using compatible consecutive measurements, not selected values from unrelated papers. All BLUESHIELD log-reduction values remain unmeasured. The WHO household-water-treatment testing protocol supplies a framework for prospective microbial challenges across organism classes and operating life [22]. The present concept has no assigned WHO performance classification.

The calculations identify unresolved design constraints: waiting time reduces delivery, membrane area depends strongly on usable flux and pressure, and UV efficacy depends on hydraulic and optical validation. These findings turn the survey's

TABLE 3. From reported survey priorities to prospective design evaluation. Percentages and counts retain the source limitations; design responses are proposed interpretations.

Reported survey evidence	Proposed design response	Prospective measurement
Feasibility favorable: 84.0%	Modular stage interfaces and service access	Assembly, leakage, operator-task and maintainability tests
Anticipated effectiveness favorable: 86.4%	Solids removal, supported membrane and terminal UV	Paired contaminant and microbial challenge measurements
Monitoring benefit favorable: 92.0%	Local sensing, flow supervision and interlocks	Calibration error, drift and alarm/valve function under faults
Environmental conditions: 20 of 25	Source-specific dosing and operating envelopes	Jar tests, UV transmittance, fouling and seasonal source variation
Energy or scalability: 15 of 25	Explicit batch timing and pressure/flow accounting	Delivered volume and electrical energy over complete operating cycles
Affordability concern: 18 of 25	Accessible consumables and service provision	Purchase, replacement, cleaning and service cost per delivered volume

The sample is concentrated in Bangladesh, students, civil engineering, and environmental science. Pooled ratings consequently reflect that composition, and no country or occupational comparison is justified without participant-level cross-tabulations and an appropriate sampling design. The reported interview counts add implementation priorities, but lack the transcripts, coding evidence, and participant quotations needed for a fully traceable qualitative interpretation. The impact-chart conflict further limits what can be concluded from that item.

Research on household treatment in Bangladesh provides a useful comparison. Luoto et al. show that assigned treatment, reported use, and stored-water microbial outcomes are distinct measurements [2]. The external results reproduced in Fig. 26 illustrate why favorable intentions should be followed by direct observation of use and water-quality measurement. Pickering et al. further support investigating user effort and service arrangements [3]. These findings motivate a future BLUESHIELD evaluation with flood-affected households and operators, rather than extending technical-stakeholder ratings to household adoption.

feasibility, energy and monitoring concerns into measurable prototype questions.

VII. DISCUSSION

The main perception items show support for evaluating BLUESHIELD, particularly its monitoring function. Their practical value lies in identifying what participants expect and which constraints deserve attention. Environmental conditions and energy concerns connect the survey to pretreatment, source-water characterization, and operating-envelope validation. Affordability concerns support evaluating recurring consumables, maintenance effort, and service availability alongside the purchase price. The survey does not quantify those costs or establish that the proposed system is affordable.

Table 3 links the reported priorities to the proposed engineering response and to measurements needed in a later study. These links are the authors' design interpretation, not causal associations estimated from the survey. They provide a practical basis for testing the concept without treating favorable ratings as evidence of treatment capability.

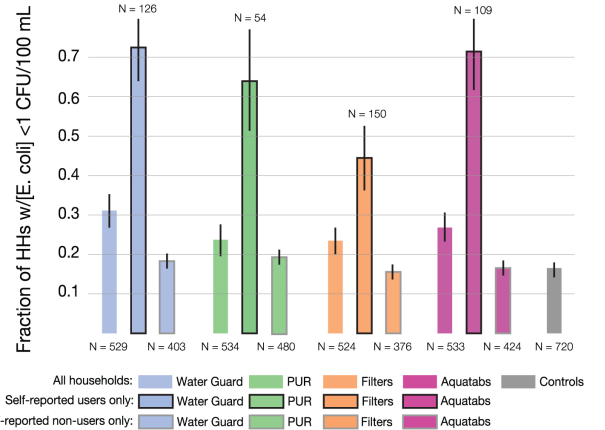


FIGURE 26. External Bangladesh field evidence: stored water with no detectable *E. coli*, by assigned product and self-reported use during the previous 24 hours. Reproduced from Luoto et al. [2], Fig. 2, © 2011 Luoto et al., under the Creative Commons Attribution License stated in [2]. Error bars are source-study standard errors. Results concern the cited products, not BLUESHIELD.

The component evidence narrows the next engineering decisions. A defined GO architecture must demonstrate stable

separation without unacceptable material release; the UV reactor must demonstrate microbial inactivation under realistic feed-water and flow conditions; and sensors must be verified as process instruments. An integrated prototype should then be evaluated through paired influent and effluent measurements, replicate operation, maintenance cycles, and storage testing. Comparisons with established treatment options are needed before claims about superiority, lower energy demand, environmental benefit, or health impact are warranted.

Seasonality also matters when planning the next survey. A longitudinal study of households in rural Bangladesh documented changes in water-insecurity experiences across the monsoon cycle [23]. A future BLUESHIELD study should therefore record recruitment season, local water sources, flood exposure, and household water responsibilities. These variables would allow the interpretation of user needs within their setting, provided that the survey instrument and sampling plan are specified before recruitment.

A. INTEGRATED MACHINE AND INDUSTRIAL INSTALLATION

The machine in Fig. 27 connects the stages in series, adds a GO feed-pressure loop and places UV-C after the membrane. Separate paths carry product, sludge and diverted water. An interlocked outlet is intended to close or divert flow during inadequate UV operation, excessive flow or other faults outside a validated envelope. Relief and diversion discharge require controlled holding, with no untreated bypass into product storage; pressure-relief settings require a rated mechanical design.

The industrial arrangement in Fig. 28 develops the earlier setup with treatment modules, residuals handling and operator access. Scaling requires source-water characterization, continuous or staggered-batch mass balances, pipe and pump sizing, redundancy, cleaning access, electrical protection and waste management. Plant capacity and flood resilience require validation. Assessing the solar-panel option requires local irradiance data, battery sizing and measured electrical demand.

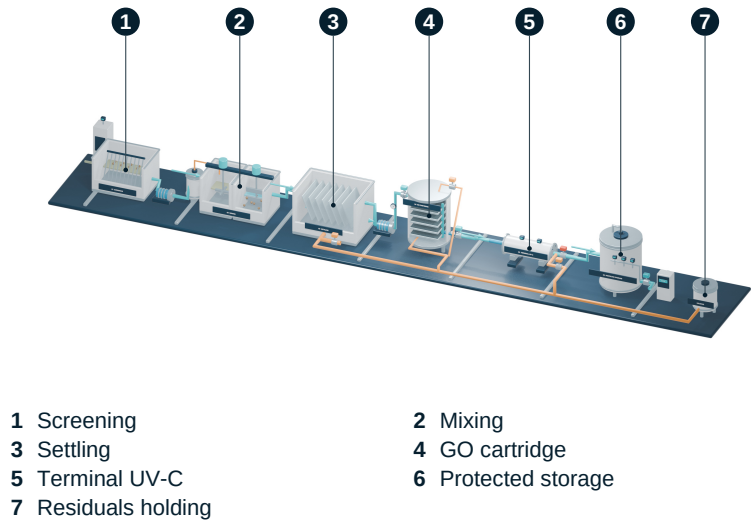


FIGURE 27. Integrated BLUESHIELD machine rendered in Blender. The revised series arrangement is intake, coagulation and flocculation, settling, supported GO, terminal UV-C and protected storage. Separate sludge and diverted-water routes and pressure-loop provisions address interfaces absent from a simple vessel diagram. Conceptual layout, without validated dimensions, capacity or performance.

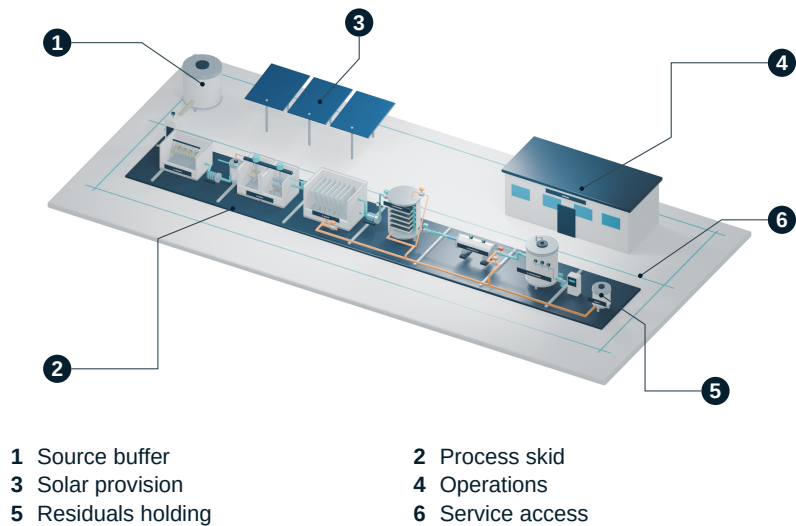


FIGURE 28. Revised industrial implementation rendered in Blender, developed from the earlier project setup. Treatment, residuals and service areas are spatially distinguished. This is a proposed installation arrangement, not an operating plant, structural design or demonstrated production capacity.

VIII. CONCLUSION

BLUESHIELD FILTER is proposed to address the treatment and operational difficulties of obtaining usable water during floods. The study combines an exploratory stakeholder survey with component evidence to define a treatment sequence, preliminary engineering requirements and priorities for prototype development. Its contribution is to connect anticipated user and operator needs with functions that can be examined through subsequent design and testing.

The proposed sequence comprises screened intake, controlled coagulation and flocculation, settling with sludge removal, a supported graphene oxide (GO) cartridge, terminal UV-C treatment and protected storage. Screening addresses coarse debris, while coagulation, flocculation and settling prepare water for the membrane. The supported GO stage provides a candidate separation platform whose function depends on material, support and operating conditions. UV-C follows filtration as the final disinfection stage before hygienic dispensing. Physicochemical sensors, flow measurement and UV-status monitoring support supervision, maintenance and fault response; appropriate laboratory measurements remain necessary to verify water quality.

The formulation integrates these treatment interfaces with practical operating provisions. Separate sludge and diverted-water routes, sampling and cleaning access, contained membrane media, a pressure-compatible feed arrangement and a proposed interlocked outlet specify the responsibilities of individual modules. The stage and installation models communicate these relationships and identify decisions needed before fabrication. This integration provides a structured basis for comparison and modification, with each proposed function linked to an observable operating condition or prospective test.

The reported survey of 125 students and professionals found favorable perceptions of technical feasibility, anticipated effectiveness and monitoring usefulness of 84.0%, 86.4% and 92.0%, respectively. In the reported 25-professional component, 20 participants identified environmental conditions, 15 identified energy efficiency or scalability, and 18 identified affordability concerns. These findings direct attention toward source-specific operation, complete-cycle energy accounting, accessible consumables and service requirements. The ratings express expectations within the recruited sample; they do not measure treatment performance or establish household adoption.

The engineering scenarios connect these priorities to measurable design choices. A 20-L batch with an assumed 90% recoverable fraction, 81 minutes of pretreatment and 10 L/h discharge yields 18 L over 3.15 hours, approximately 5.71 L/h across the cycle. The recoverable fraction is a lumped volume

allowance; filling, cleaning and operator delays are omitted, so this calculation is not a continuous production rating. At assumed fluxes of 10, 25 and 50 L m⁻² h⁻¹, delivering 10 L/h requires 1.0, 0.4 and 0.2 m² of membrane area. Pressure and UV-exposure calculations identify measurements required to establish sizing and performance.

Recruitment procedures, the complete questionnaire, collection dates, participant responses and interview transcripts are unavailable, and the sample is concentrated in Bangladesh and selected disciplines. Conflicting impact-item summaries remain excluded from interpretation. Consent and ethics documentation have not been recovered, preventing verification of the original collection procedures. These limitations restrict reproducibility and generalizability, while absent treatment experiments leave contaminant removal, potability, operating cost and sustained performance unresolved.

Further work should combine documented survey expansion with staged prototype evaluation. Recruitment toward at least 350 participants should follow a justified sampling and study-size plan, include intended users and operators, and record regional and seasonal context. A larger count alone will not correct selection bias. Prototype testing should establish source-water envelopes through jar tests, characterize the supported GO architecture, and measure fouling, material release, permeate quality and cleaning performance. UV exposure and microbial inactivation, sensor calibration, outlet interlocks and storage hygiene require evaluation. Modifications to membrane area, pressure supply, batch scheduling, replaceable modules and power provision should follow measured throughput, energy, maintenance and cost data. Comparison with established options and observation of sustained household use would then determine where the concept offers practical benefit. BLUESHIELD presently provides a survey-informed, testable design direction for floodwater treatment.

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